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Developments of a Method for Lap Time Simulation

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University of Brescia

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Developments of a Method for Lap Time Simulation

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ABSTRACT

The paper aims at introducing the developments of a performance simulation method for predicting the potential lap time of a given racing car on a given circuit. It is based on a previous work published in [1].

This first attempt produced fairly reliable results but it was not totally satisfactory since computation was very slow. It was then decided to restart from scratch with the following approach. Reference is made to the renowned «Milliken» [2].

The vehicle maximum G-G envelope is estimated. Then the model follows the «real» raceline -acquired with on-board instrumentation- performing within the G-G envelope. Accurate comparison with real-world data is used to identify and calibrate some of the model parameters. After calibration the model can be used to study the effect of different set-ups, aerodynamic balance, gear ratios etc. Where the circuit is new to the team a trajectory can be designed on a circuit map by using a built-in, parametric CAD-like interface.

G-G envelope estimation prior to actual simulation has largely improved the computational speed. Also, the graphic interface aims at pit-lane use by the race engineer.

The software is the product of a joint research between the University of Brescia and the team BMS Scuderia Italia therefore it is not available on the market.

INTRODUCTION

The application of Vehicle Dynamics to the world of racing is relatively simple. Unlike road vehicles, racing cars aim at one main target: the best possible lap time. Therefore, the prediction of lap times and the related computer simulation is virtually one of the most useful software tools for any race engineer.

The concept is not really innovative: as stated in [2] Mercedes was the first in undertaking lap time computation when computers didn't even exist i.e. in the mid fifties. In this case the main targets were:

- high computational speed, easiness and versatility for «pit-lane use» by the race engineer
- interaction with data acquisition for estimating parameters like tire friction level, aerodynamic forces etc and validating the simulation
- simple modeling of driver's skill
- computation of the raceline on an unknown circuit
- supporting the design of a new circuit

VEHICLE MODELLING

The software should enable the race engineer to estimate the effect of various setup settings in the shortest possible time. Therefore a high computational speed was absolutely one of the main targets.

A trimmed steady-state analysis is used (see [2]) to define the maximum performance envelope of the vehicle as a G-G diagram. This technique is based on the assumption that the vehicle inertia moment is negligible compared to the yaw moment developed by the tires, that is to say transient cornering phases can be neglected and driving is considered as smooth switching from one trimmed situation to the following one.

The vehicle model is basically as described in [1]; suspension kinematics, vertical and roll stiffness and aerodynamic parameters are given as linear coefficients, while tires are a combined model based on Prof. Pacejka's Magic Formula (see Figures 1, 2, 3). Anyway the G-G diagram can be computed with more complete models as well, provided the trimmed steady-state technique is used. A "step 2" vehicle model with non-linear suspension stiffness and aerodynamic maps as obtained by advanced wind tunnel testing is currently under development.

The model is subjected to a full range of combined lateral and longitudinal accelerations to find the performance limits regardless of its final behavior. The

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G-G diagram is then stored as a function of speed i.e. as a 3D surface (Figure 4).

BEHAVIOR EVALUATION

Various forms of behavior evaluation are available to the user.

The G-G diagram can be integrated with an understeer gradient map (Figure 5). Moreover, since different drivers can tolerate different levels of under/oversteer the diagram can be cut off to suit their capability (Figure 6). Also the driver's braking reaction is adjustable for this purpose.

Also the understeer gradient can be traced vs. lateral acceleration and speed (Figure 7).

Any point in the diagram can be clearly interpreted with the use of a typical «friction circle» (Figure 8).

ENGINE AND TRANSMISSION

The G-G diagram is computed for an ideal car with unlimited power; i.e. the performance is not affected by engine and driveline characteristics. Only at the simulation stage power curve, efficiency and gear ratios are considered. This makes gear ratio search and optimization much quicker since there is no need for computing the maneuvering envelope each time (Figure 9).

VALIDATION, TUNING, SIMULATION...

The validation is obviously based on the interaction with on-board logged data. The vehicle model is run on the real raceline, where radius $R = V^2/Ay$.

Real world and simulated data are then superimposed for a direct comparison (Figure 10). Vehicle parameters can be adapted or identified with a suggested procedure with the aim of reproducing real world performance with the best possible precision.

...AND POST-PROCESSING

From this stage onwards the model can be used to estimate the effect of setup variations. It is particularly useful when searching for the best aerodynamic compromise, provided reliable data are available.

As stated in the introduction the program is intended for use as a support to data acquisition analysis in the pits rather then as a speculative tool for the office desktop. Therefore a big effort has been put in the creation of an intuitive and user-friendly interface. Various post-processing options are available as shown in the pictures; a link is active between the output history vs time/space and performance charts like the G-G diagram or the friction circle plot (see Figure 11).

RACELINE AND CIRCUIT DESIGN

Where a circuit is "unknown" and no experience neither reliable telemetry data are available, usually general balance and gear ratios are chosen on a rough analysis of the circuit configuration.

This is even more problematic when a brand new circuit project is under way. Cornering speed must be estimated to determine safety characteristics, crash barriers and runaways. Also, the design is usually aimed at the best possible compromise between circuit length and average speed within the available area lot.

One of the program modules is conceived to estimate a raceline based on a raster image of a "new" or unknown circuit. A parametric algorithm and a CAD-like interface (Figure 12) enable the design of a sensible raceline according to drivers' and engineers' experience. Altimetry can be taken into account as well where it does affect the overall performance.

CONCLUSION AND FUTURE DEVELOPMENTS

A method for predicting the lap time of a given racing car on a given circuit has been described. This is based on previous research [1] as well as on references like [2] and [3].

Such method has originated a software -still under development- conceived as a support to data acquisition.

As underlined a more comprehensive vehicle model focused on non-linear aero maps and suspension rates is in progress. Another promising development is a stronger integration with the data acquisition software to speed up validation and simulation and compensate for some post-processing gaps.

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2. Milliken W.F., Milliken D.L.: «Race Car Vehicle Dynamics». SAE, 1994.
3. Guiggiani M.: «Dinamica del Veicolo». UTET, 1999.

CONTACT

Marco Gadola is currently researcher at the University of Brescia. His race engineering experience includes Le Mans, GT and sports prototypes championships.

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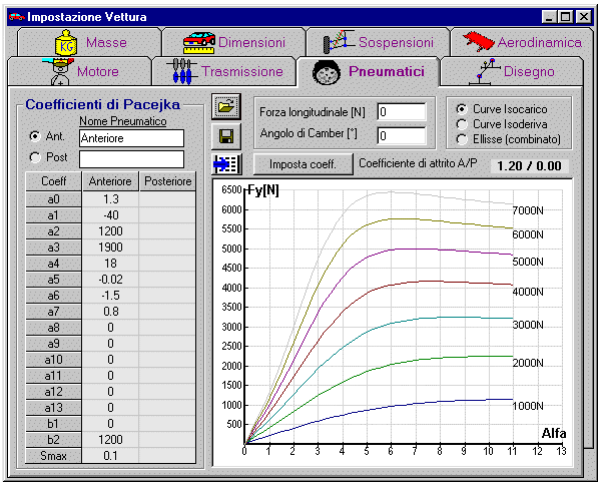

Figure 1 – Pacejka's tire performance curves

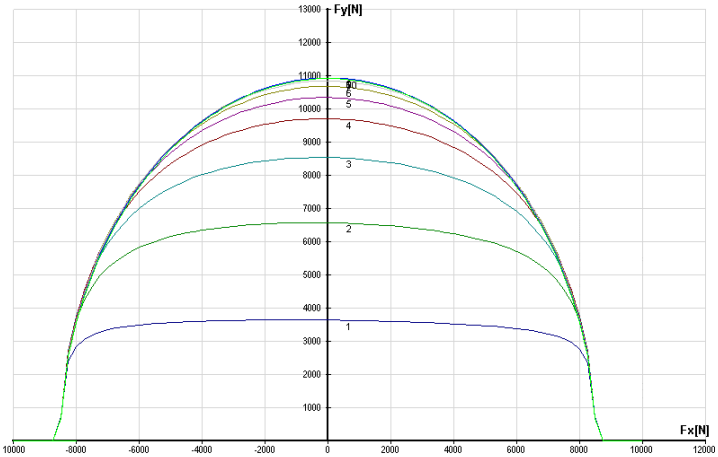

Figure 2 – Combined lateral/longitudinal tire curves

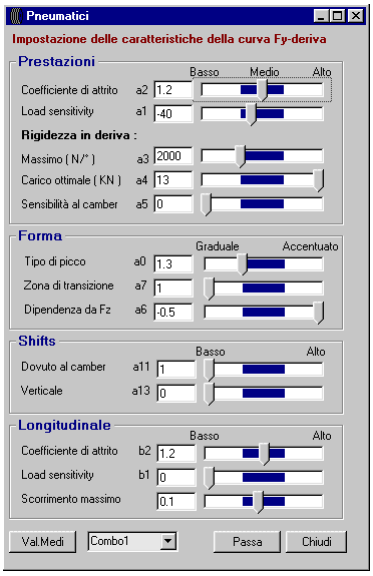

Figure 3 – Pacejka's coefficient control panel

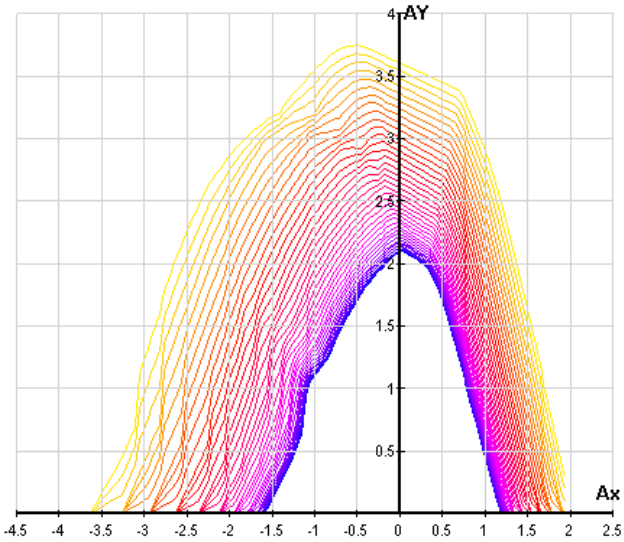

Figure 4 –G-G diagram vs speed

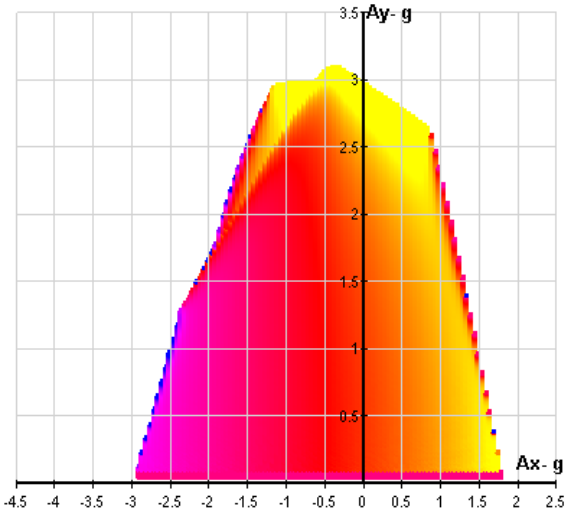

Figure 5 – G-G diagram vs understeer gradient

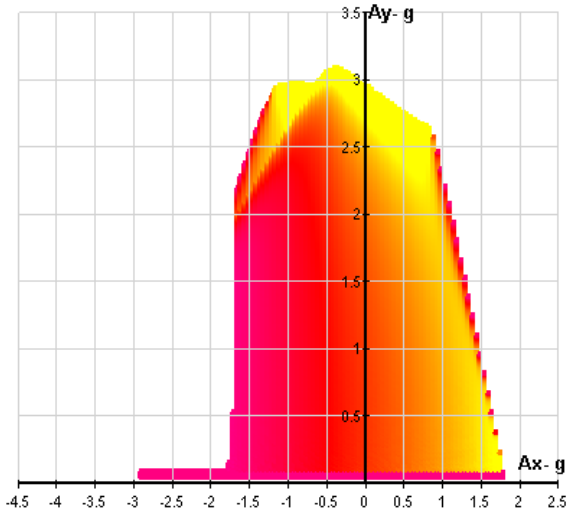

Figure 6 – reduced G-G diagram

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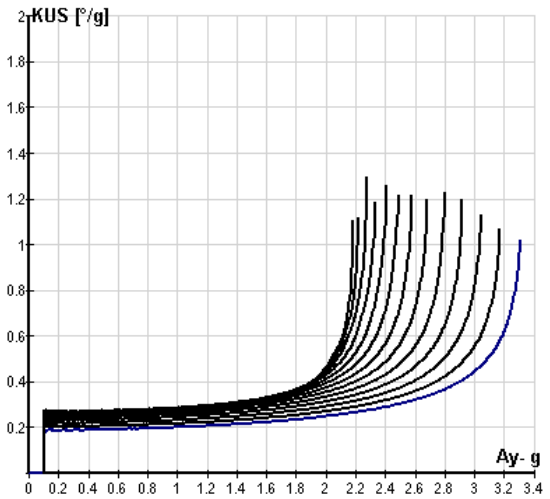

Figure 7 – understeer gradient vs lateral acceleration

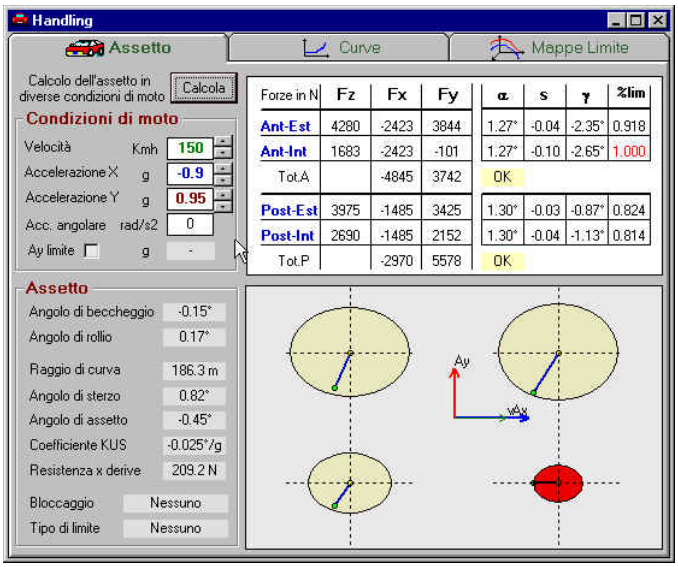

Figure 8 – friction circle window

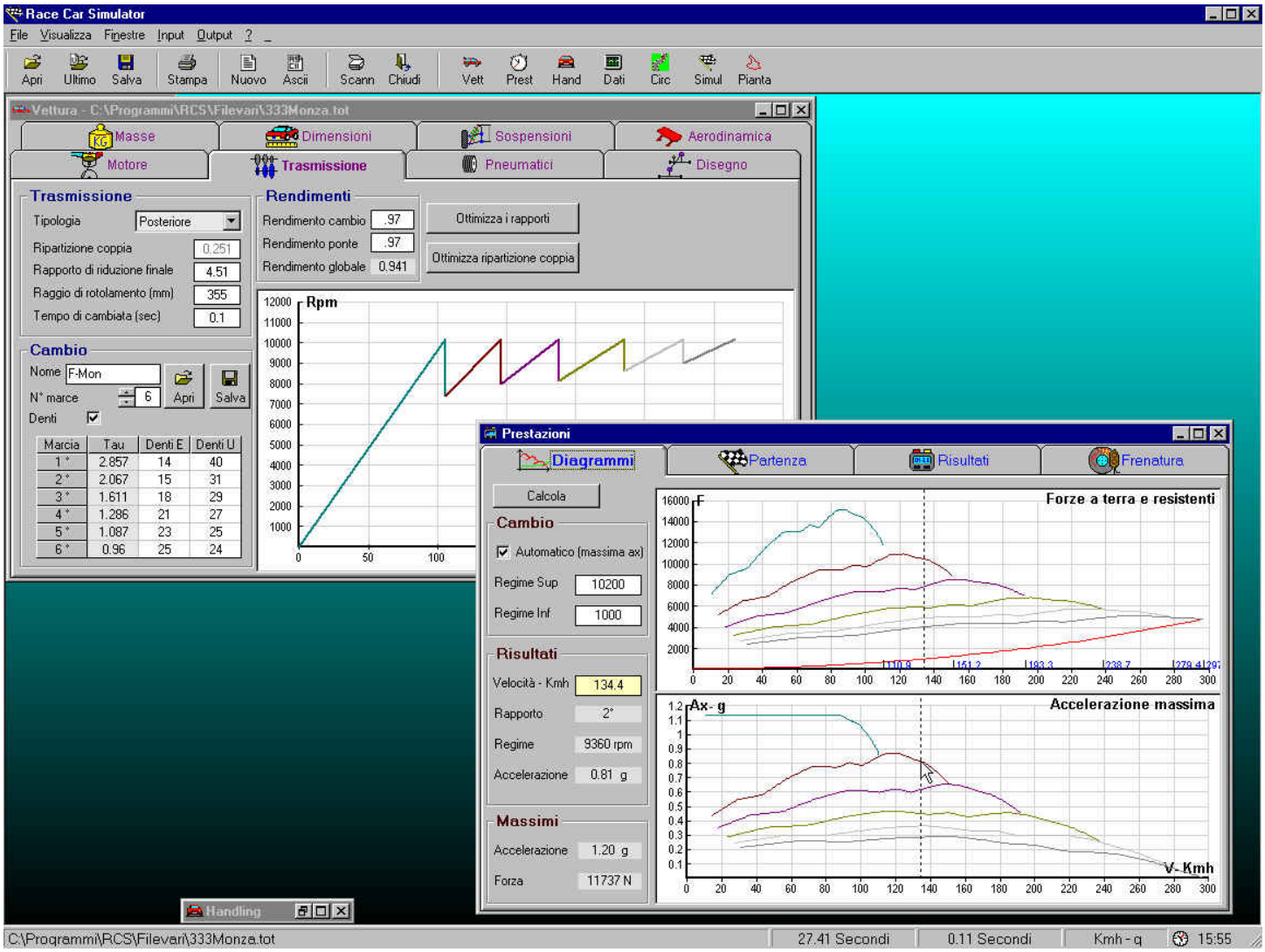

Figure 9 – Transmission and performance windows

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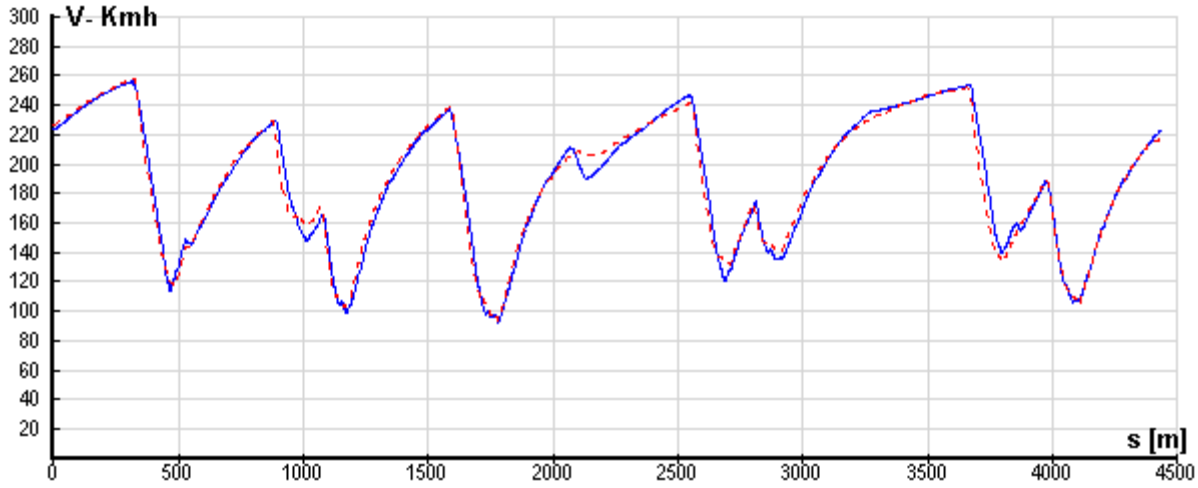

Figure 10 – Real vs simulated speed (Nurburgring)

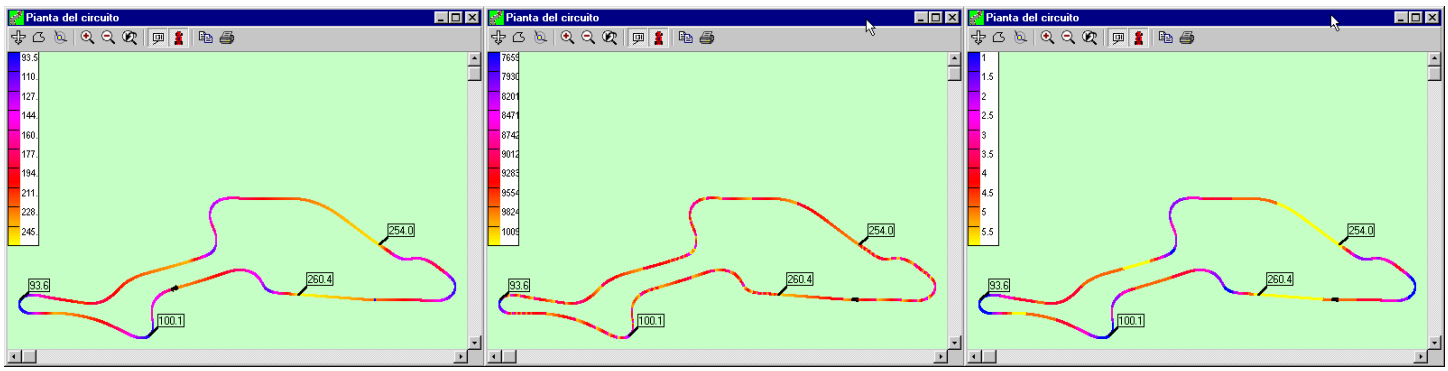

Figure 11 – Speed, rpm and gear on the Nurburgring

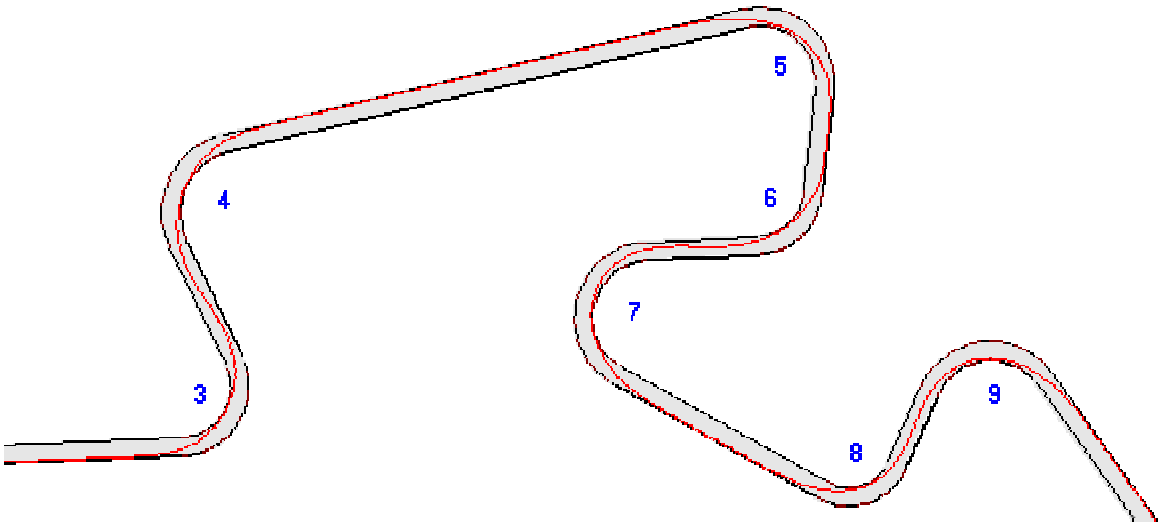

Figure 12 – Racing line design for an unknown circuit